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Evaluating the effectiveness of social media advertising strategies across different demographics and platforms

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ABSTRACT

This paper aims to evaluate the effects of demographic targeting, platform selection, and engagement metrics on the effectiveness of social media advertising campaigns. In this paper, it is also aimed to provide a comprehensive understanding of how these factors influence conversion outcomes. We employed a hybrid analytics method combining ANOVA, correlation, and regression analysis to examine patterns of user engagement and ad performance. A dataset of 200,235 ad campaign records conducted in the United States was analyzed to assess variations across gender, age groups, and platforms. The main findings of the study demonstrate that (i) women consistently exhibited higher conversion rates compared to men, highlighting the potential of gender-specific content strategies; (ii) users aged 35–44 achieved the highest engagement, suggesting that middle-aged audiences may be more responsive to targeted campaigns; and finally, (iii) among platforms, Instagram led in conversion performance, while Facebook showed comparable engagement rates, and both significantly outperformed other platforms in terms of overall impact. Click-through rate was found to be moderately correlated with conversions, whereas impressions showed weak returns, indicating diminishing value at higher exposure levels. These results underscore the importance of optimizing both message and delivery channel to specific audience segments.This study contributes to digital marketing literature by integrating demographic variables, engagement indicators, and platform performance into a unified model. The insights derived offer practical implications for businesses and advertisers aiming to enhance return on investment and optimize campaign strategies in a fast-paced, digitally driven advertising landscape.

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Introduction

Social media has become an indispensable element of modern advertising, enabling brands to engage with targeted audiences in ways that were unimaginable just a decade ago. As platforms like Facebook, Instagram, and TikTok reshape the marketing landscape, advertisers now have access to an abundance of tools for micro-targeting and real-time feedback. However, this evolution introduces complexity: the effectiveness of advertising campaigns on social media hinges not only on the platform and its features but also on the demographic characteristics of the audience and the campaign's ability to foster engagement. In a time where billions are spent on digital advertising each year, understanding the mechanics behind successful campaigns has never been more urgent.

Despite the widespread use of social media as a marketing channel, scholarly understanding of how demographic variables, such as age and gender, interact with platform-specific attributes to influence conversion outcomes remains limited (Dwivedi et al., 2021; Choudhary et al., 2024; Hudders & De Jans, 2022). While younger users are often presumed to be more responsive to digital content, recent studies suggest more nuanced behaviors across generations (Roberts & David, 2019; Zoghi et al., 2024). Similarly, the relationship between common engagement metrics (e.g., impressions, clicks, likes) and actual ROI or conversions is often overstated, with some scholars warning against overreliance on superficial indicators (Hollebeek & Macky, 2019; Hirsch, 2014). Furthermore, research tends to isolate platform analysis rather than exploring cross-platform performance differences in a unified empirical

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framework (Rosli et al., 2024; Shen, 2023). These gaps reflect the need for an integrated, data-driven approach that can guide advertisers in making more strategic decisions.

This paper aims to evaluate the effects of demographic targeting, platform selection, and engagement metrics on the performance of social media advertising campaigns. Specifically, it seeks to examine how age and gender influence engagement and conversion outcomes; how platforms differ in effectiveness; and how key engagement metrics predict conversion rates. We employed a quantitative methodology using a large-scale dataset of 200,235 samples from ad campaign records drawn from various platforms and demographic groups in the United States, applying statistical tools including ANOVA, correlation, t-tests, and regression analysis.

This paper is organized as follows: following the introduction, the second part provides a review of theoretical and empirical literature exploring the linkage between digital advertising strategies and consumer behavior. The third section introduces the research background and methodology. This is followed by the presentation of analysis and key findings. The discussion section interprets the results in light of existing literature, and the final section concludes with the study's main contributions, practical implications, limitations, and suggestions for future research.

Literature Review

The purpose of this literature review is to explore how social media advertising effectiveness varies across platforms and demographic groups, with a particular focus on engagement metrics and conversion outcomes. As advertising continues to shift toward digital spaces, understanding platform-specific dynamics and audience behaviors becomes critical for optimizing marketing strategies. This review synthesizes recent empirical studies and theoretical insights to identify patterns, contradictions, and gaps in current research. A comparative and integrative approach was adopted, emphasizing cross-platform differences, generational and gender-based audience responses, and the impact of various advertisement formats and features. By highlighting both established findings and areas in need of further exploration, such as the effects of newer platforms like TikTok and the tension between personalization and privacy, this review lays the foundation for a more data-driven and targeted advertising strategy in future research and practice.

Platform differences in social media advertising effectiveness

Studies consistently indicate that the impact of an advertisement is quite different from one social media platform to another. This is partially explained by the variation in user base, nature/features of platform, and trust of users. The acceptance of ads seems to be determined by user trust and reliability with a platform. For example, Nuji et al. (2023) discovered that advertisements on highly credible, professionally focused websites (e.g., LinkedIn) are likely to be viewed as more credible than advertisements on less moderated sites. Credibility, in turn, has been connected with more engaging consumer experience and greater purchasing intent. This implies that a website's credibility and content moderation approach can influence the persuasiveness of its advertising. Marketers can thus anticipate better performance on sites that users inherently trust for content quality. User interaction activities also differ from site to site, affecting ad performance.

Platform design and advertisement form also contribute to efficacy. For instance, Rosli et al. (2024) found that Instagram's "like" and "share" features were more strongly associated with purchase intention than those on Facebook, suggesting that visual-first platforms may foster more action-oriented engagement. Their study underscores the importance of not only measuring engagement volume but also understanding the type of engagement elicited by specific platform features. Shen (2023) discovered that brand engagement can be enhanced through the selection of a suitable platform strategy, rather than adopting several content strategies for each platform, which corroborates the evidence that amidst the rapid evolution of social media, its sheer variety makes it hard for brands and marketers to concentrate on all the platforms (Gawer, 2022). In Pelet and Ettis's (2022) research on Facebook advertisements, it has been discovered that some attributes of social media advertisements: creativity, entertainment ("liking"), and credibility exert a positive impact on customers' attitudes towards the advertisement and thereby induce purchase intention.

Intrusive or irritating formats, on the other hand, can harm ad effectiveness. Comparative studies across media have been comparatively rare, but recent research is closing this gap. Yang et al. (2024) provided empirical support for the ambitious engagement-to-conversion promise of TikTok by illustrating the effect of influencer video features on user engagement. The findings indicated that entertainment and informativeness were significant predictors of behavioral intention.

Demographic targeting and audience differences

The general consensus through literature is that various demographic segments use social media and advertising differently and thus need to be addressed through different strategies (Dwivedi et al., 2021). One core factor is age or generation groups. It has been discovered through previous research that younger users (Gen Z and Millenials) are digital natives and use social platforms excessively, while older groups (such as Gen X and Boomers) access them less and for other reasons. These patterns of use translate to several different advertising outcomes. Interestingly, there have been studies questioning traditional assumptions about what age groups are most susceptible to social advertisements. Choudhary et al. (2024) conducted a systematic review of generational differences in technology-related behavior, which includes insights into how various age groups engage with social media and digital advertising. Their findings suggest that while both younger and older users are exposed to social media ads, younger generations,

particularly digital natives like Gen Z, tend to exhibit higher engagement and purchasing behaviors. This challenges assumptions that older individuals are more susceptible due to lower digital literacy. In contrast, the review indicates that frequent social media exposure, along with generational traits such as FOMO and peer influence, may actually make younger users more responsive to advertising content. In reality, younger audiences, as social media savvy as they are, were more vulnerable or ready to purchase from social media platforms. This paradox underscores the point that increased exposure and familiarity with social media can also imply increased advertisement effectiveness in terms of conversions. It also suggests an absence of understanding of psychological motivators by age: while young users are aware of adverts, they might also be more prone to FOMO (Fear of missing out) as stated by Roberts and David (2019) or social influence that triggers purchases, something to be looked into further (as Khan et al., 2020 explain).

Gendered differences in response to social media advertising are another key sub-theme.

Recent research points to nuanced differences in how males and females interact with and trust ads or influencers online. Hudders and De Jans (2022) described that females are more drawn to visual-intensive, entertainment-based advertising material (common on platforms like Instagram) than men, leading to higher interaction with these kinds of advertisements. Particularly, women on Instagram were most interested in advertisements that are vivid and story-like compared to men (Hudders & De Jans, 2022). Additionally, Khan (2020) examined social media influencer marketing and found a gender moderator role: the female consumer and male consumer differed in the conversion of influencer endorsements to purchase intention. Female respondents were more influenced by social media influencers (specifically authentic micro-influencers) in terms of purchase intent, whereas male respondents were less so. This aligns with the view that women are more likely to trust peer recommendations and engaging social content, in line with findings that women value authenticity and two-way dialogue with brands on social media. Men, however, might be more likely to react to other appeals – e.g., there is evidence that men are more motivated by functional and information-driven content in advertising, with women being more reactive to experiential and emotional appeals (Peña-García et al., 2018).

Engagement metrics

Social media campaigns are often evaluated on likes, shares, comments, click-through rates, etc., but the relationship between these metrics for engagement and actual conversion or sales is not fully clear.

Some reports only provide weak or indirect correlation, e.g., Hollebeek and Macky (2019) argue that high online engagement may not necessarily translate into a high growth in sales or loyalty because there are too many intervening variables.

The experimental research of Hirsch (2014) highlighted this gap too: they did not only quantify online behavior, but also offline actions of behavior (e.g. donations) and discovered some calls to action could cause clicks without contributing to actual meaningful activities.

Contradictions and research gaps

In terms of gaps, academic research has only begun to keep pace with the rapid evolution of social media platforms.

Firstly, cross-platform comparative studies are scarce. Many studies focus on one platform at a time (often Facebook or Instagram), leaving a gap in understanding how the same advertisement or strategy performs across different platforms under controlled conditions. There is a need for studies that examine platform-specific advertising effectiveness, considering demographics and engagement behavior, as well as the impact of newer formats and algorithms.

One contradiction lies in generational responses: earlier assumptions held that the digitally native Gen Z generation would be desensitized and difficult to persuade, yet recent evidence shows Gen Z and Millennials are in fact more likely to trust and act on social media ads than older adults.

Second, emerging platforms like TikTok (short-form video) or features like Stories and Reels on Facebook introduce new creative possibilities, but their long-term marketing effectiveness is not yet well documented in scholarly literature (most data on these come from industry reports).

Another area of mixed findings is personalization and privacy: personalized targeting is attempting to improve ad relevance, but some studies (e.g. Balaban et al., 2022) note that overly personalized ads can trigger privacy concerns and backfire for certain users. The balance between effective targeting and consumer comfort remains an open question, especially as regulations and user awareness of data privacy grow.

Addressing these gaps will be crucial for a holistic understanding of social media advertising effectiveness. This literature further justifies the relevance of the topic, by stressing the fact that social media advertising strategies should be multifaceted, in order to be more efficient. This study seeks to close the gaps in terms of cross-platform comparative studies and provide useful recommendations to advertisers by looking at demographics, platforms and conversion rates, all within one analysis. Thus the results will enhance the effectiveness of the multi-channel marketing strategy, by improving quantitative characteristics of advertising campaigns aimed at different audience markets.

Research question / Hypothesis

The research paper was based on the following research questions paired with the hypotheses designed to explore what are the factors influencing the performance of social media advertising campaigns. They draw on existing literature and have an aim to examine the relationships between demographic factors, platform characteristics, engagement metrics, and conversion rates, which will offer further insights into how these variables have an impact on the success of targeted campaigns.

How do demographic factors (age, gender) influence the performance of social media advertising campaigns in terms of engagement, conversion rates, and ROI?

This question examines the role that age and gender have in the performance of media advertising campaigns in terms of metrics that could point to successful campaigns.

How do social media platforms (Facebook, Instagram, X) differ in their ability to drive conversions and engagement for targeted demographics?

This question explores how effective are different social media platforms in reaching specific demographic groups and driving desired outcomes.

What is the relationship between engagement metrics (impressions, clicks, engagement score) and conversion rates across different platforms?

This question investigates the correlation between key engagement metrics and actual conversion outcomes, a focus being put on understanding how engagement drives conversions on different platforms.

H1.1: Campaigns targeting younger demographics (18–24 years) will have higher engagement scores than campaigns targeting older demographics.

Previous research suggests that younger audiences are more active on social media platforms, engaging more with content, especially on platforms like Instagram and TikTok, which cater to their preferences for visual and interactive content.

H1.2.: Campaigns targeting female audiences will have higher conversion rates compared to those targeting male audiences.

Gender differences in content preference and response have been documented, with women often engaging more with visually appealing ads and social media content, leading to higher conversion rates for campaigns targeting them.

H2.1: Engagement rates will be significantly higher on Instagram compared to Facebook.

Instagram's visual-centric platform is more suited to campaigns that emphasize aesthetics and imagery, leading to higher engagement rates for campaigns focused on visual content, especially among younger audiences.

H2.2: Conversion rates will be higher on Facebook than on Instagram for campaigns targeting older demographics.

Facebook's broad and diverse user base, which includes a significant proportion of older adults, may lead to higher conversion rates for campaigns targeting this demographic because Facebook has existed for a longer time, and has accumulated a large number of users that fit in the older demographic, while not drastically changing their platform and functions.

H3.1: Higher click-through rates (CTR) will be positively correlated with higher conversion rates.

Studies suggest that a higher CTR indicates a strong initial interest in the ad, which is often a precursor to conversions. With this said, ads with a higher CTR are expected to have higher conversion rates, as they indicate that the audience is more likely to act on the ad's call-to-action.

H3.2: Campaigns with higher impression counts will lead to diminishing returns on conversion rates beyond a certain threshold.

While increasing impressions can raise awareness and drive engagement, research has shown that beyond a certain point, additional impressions may lead to oversaturation, reducing their effectiveness in converting users into customers. This reflects the principle of diminishing returns, where the cost of additional impressions outweighs their benefit in terms of conversion.

Research and Methodology

The methodology of this study is based on the analysis of an authentic, pre-existing data set from social media advertising campaigns to ensure real-world applicability and validity. The research uses a quantitative approach, using statistical methods to test the proposed hypotheses and explore the relationships between key variables. It uses a correlational research design that has an aim to explore the relationships between platform features, demographic factors, engagement factors and conversion rates. The sample consists from data of advertising campaigns from various industries that target different demographic groups.

The data set* contains the following variables:

Acquisition cost, campaign goal, campaign id, channel used, clicks, company, conversion rate, customer segment, date, duration, engagement score, impressions, language, location, ROI, target audience.

Demographics of the target audience: Age, gender, and city.

Platform: Facebook, Instagram, X (further mentioned as Twitter) and Pinterest.

Engagement Metrics: Impressions, clicks, engagement score.

Conversion Data: Conversion rates, ROI.

Campaign Goals: Product launch, brand awareness, market expansion, increase sales

Language of the campaign: English, Spanish, French.

Customer Acquisition Cost and **Duration of campaign.**

*The data set is publicly available and can be found on the following website: https://www.kaggle.com/ .

Even though the data set contained an array of data, only several variables, the ones that were deemed to be useful to provide comprehensive research, were used in the analysis. The data used was from the United States of America.

The independent variables are the demographic factors, campaign goals and platform, while the dependent variables are engagement metrics, conversion data and customer acquisition cost.

To test the hypotheses, an array of statistical methods were used, conducted through the statistical software SPSS:

Descriptive analysis summarized the key features of the data, including averages, standard deviations, and frequency distributions, providing an overview of the sample and the distribution of the key variables.

Pearson Correlation was used to examine the relationships between engagement metrics and conversion rates, helping determine whether higher engagement correlates with higher conversions across different platforms.

T-tests were used to compare the engagement and conversion rates between gender groups and demographic age groups.

ANOVA was applied to compare engagement and conversion rates across multiple platform categories and campaign goals.

Multiple linear regression analysis was employed to explore the relationship between engagement metrics and conversion rates, controlling for demographic variables and platform characteristics. This allowed for a deeper understanding of how engagement metrics predict conversion outcomes across different platforms and demographic segments.

Findings and Discussion

This section provides a summary of the data set, including key statistics for numerical variables and frequency distributions for categorical variables. The analysis helps understand the data set's composition before conducting further statistical tests.

Descriptive statistics for numerical variables

The table below presents the mean, median, maximum, minimum and standard deviation of values for key numerical variables: acquisition cost, clicks, conversion rate, date, duration, engagement score, impressions, ROI.

Table 1: Descriptive Statistics

Descriptive Statistics					
	N	Minimum	Maximum	Mean	Std. Deviation
Acquisition_Cost	200235	1000	14999	8000.27	4043.117
Clicks	200235	173.16000000000000	115177.60100000000000	26553.604432792830000	25472.115375226140000
Conversion_Rate	200235	.0023710579199999	1.6682484961592614	.160116803594289	.154938543157617
Duration	200235	1	30	15.45	8.651
Engagement_Score	200235	2.0670000000000000	921.5692800000000000	239.776407023014830	199.218156412293900
Impressions	200235	620.1000000000000	276470.7840000000000	71932.922106904230000	59765.446923688745000
ROI	200235	.000006162424385	127.165922909019640	13.464683300344786	13.049559568431407
Valid N (listwise)	200235				

Acquisition Costs

The average acquisition cost per campaign is 8,000.27 (SD = 1,043.12), with values ranging from 1,000.00 to 14,999.00. The relatively high standard deviation suggests considerable variation in cost allocation across campaigns.

Click performance

Campaigns generate an average of 26,553.60 clicks, though the median value is significantly lower at 17,513.78, highlighting a skewed distribution. The standard deviation (26,202.50) further indicates substantial differences in user engagement across campaigns.

Conversion rates

The mean conversion rate is 16.01%, with a minimum of 2.37% and a maximum of 116.67%. The substantial variation suggests that some campaigns outperform others significantly in terms of converting clicks into meaningful actions.

Campaign Duration

The typical campaign runs for 15.45 days on average, with a minimum duration of 1 day and a maximum of 30 days. These variations could reflect differences in advertising strategies and budget allocations.

Engagement metrics

Engagement scores vary widely, with a mean value of 239.78 (SD = 219.18). The highest engagement score recorded is 921.57, while the lowest is 2.07, indicating significant disparities in audience interaction.

Impressions and reach

Campaigns generate an average of 71,932.92 impressions, with some campaigns reaching up to 276,470 users. The high standard deviation (59,765.45) underscores the substantial variability in campaign reach.

Return on Investment (ROI)

The mean ROI is 13.46, but the presence of extreme values (ranging from 0.00006 to 127.17) suggests that certain campaigns yield disproportionately high returns. The standard deviation (34.91) further highlights the inconsistency in campaign profitability.

Statistical observations

Standard deviations for key performance indicators (KPIs) such as clicks, engagement, and ROI suggest that campaign performance differs widely.

Skewness and kurtosis values indicate that most variables exhibit non-normal distributions, which is further confirmed by the Jarque-Bera test ($p < 0.001$).

Certain campaigns achieve significantly higher engagement and ROI, warranting further analysis into factors contributing to their success.

Frequency distribution for categorical variables

Categorical values that will be included in the analysis from the data set are Target audience (a new variable, "Gender" was created) and Channel used (further referred to as "Platform"). Other categorical variables are Campaign goal, Location, Language, Company and Customer segment, which will not be used in this analysis.

Frequency distribution of "Channel used"

Table 2: Frequency Distribution of "Channel Used"

Channel_Used					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Facebook	65786	32.9	32.9	32.9
	Instagram	60077	30.0	30.0	62.9
	Pinterest	30030	15.0	15.0	77.9
	Twitter	44342	22.1	22.1	100.0
	Total	200235	100.0	100.0	

To assess the distribution of advertising campaigns across different social media platforms, a frequency analysis was conducted on the categorical variables "channel used". The data set comprises 200,235 observations distributed across four platforms:

i. Facebook: 32.85% (65,786 campaigns)
ii. Instagram: 30.00% (60,077 campaigns)
iii. Pinterest: 15.00% (30,030 campaigns)
iv. Twitter: 22.14% (44,342 campaigns)

The results indicate that Facebook and Instagram are the most frequently used advertising platforms, collectively accounting for over 62% of all campaigns. Pinterest represents the lowest proportion at 15%, while Twitter accounts for 22.14% of the campaigns. The preference for Facebook and Instagram suggests that advertisers prioritize these platforms, likely due to their larger user bases and advanced ad-targeting capabilities.

Frequency distribution of "Target audience"

Table 3: Frequency Distribution of "Target Audience"

Target_Audience					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	All Ages	17838	8.9	8.9	8.9
	Men 18-24	22399	11.2	11.2	20.1
	Men 25-34	22571	11.3	11.3	31.4
	Men 35-44	25059	12.5	12.5	43.9
	Men 45-60	15299	7.6	7.6	51.5
	Women 18-24	24303	12.1	12.1	63.7
	Women 25-34	33879	16.9	16.9	80.6
	Women 35-44	24920	12.4	12.4	93.0
	Women 45-60	13967	7.0	7.0	100.0
	Total	200235	100.0	100.0	

The frequency analysis of the categorical variable "target audience" shows the frequency of the age and gender distribution of the data set. The sample is composed of 200235 individuals.

The highest percent of users are of the category "Women 25-34" with 16.92% (33879 individuals), and the second highest being "Men 35-44" with 12.51% (25059). The lowest percentage of users are of the category "Women 45-60", with 6.98% (13967).

The "Men 18-24" and "Women 18-24" categories are also notably represented, with 11.19% and 12.14% of the sample, respectively, suggesting strong engagement from younger age groups.

The distribution appears to be somewhat skewed toward younger and middle-aged adults. More than 60% of the sample consists of individuals aged 18-44, with only a small portion (less than 15%) represented by those aged 45 and older. This suggests that younger demographics may be more engaged with the social media platforms represented in the data set.

This skewness could reflect broader trends in social media usage, where younger audiences are more likely to use a platform like Instagram, which might not be as appealing to older adults. This can help explain the overrepresentation of younger adults and the underrepresentation of older demographics.

Additional to this, a new variable was created named "Gender":

Table 4: Frequency Distribution of "Gender"

Gender					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	103166	51.5	51.5	51.5
	Female	97069	48.5	48.5	100.0
	Total	200235	100.0	100.0	

The study sample consisted of a total of 200,235 participants, with a slightly higher proportion of males (103,166; 51.5%) compared to females (97,069; 48.5%). This near-equal gender distribution suggests a balanced representation of both groups, reducing potential gender-based sampling bias. While the difference in numbers is relatively small, it may still be relevant when analyzing trends and engagement patterns across gender categories. Further examination of gender-based differences could provide deeper insights into potential variations in behavior, preferences, and response rates within the study.

Hypothesis testing

How do demographic factors (age, gender) influence the performance of social media advertising campaigns in terms of engagement, conversion rates, and ROI?

This section presents the results of hypothesis testing for H1, which examines the impact of demographic factors age and gender on engagement and conversion rates.

H1.1: Engagement scores will be higher for younger age groups compared to older age groups.

One-Way ANOVA: Age Group & Engagement Score

Table 5: One-Way ANOVA for "Engagement Score"

ANOVA					
Engagement_Score					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	796092.110	4	199023.028	5.015	<.001
Within Groups	7946065639.232	200230	39684.691		
Total	7946861731.342	200234			

Key Findings:

F(4, 200230) = 5.015, p = .000 → The p-value is significant (< 0.05), meaning age group has a statistically significant impact on engagement score.

Between-group variance (Sum of Squares = 796,092.110) shows that differences exist among age groups.

Post Hoc Test (Tukey HSD)

Table 6: Post Hoc Test for "Engagement Score"

Multiple Comparisons						
Dependent Variable: Engagement_Score						
Tukey HSD						
(I) Age_Group	(J) Age_Group	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
18-24	25-34	-.048952766007659	1.246093452340081	1.000	-3.448045540035533	3.350140008020215
	35-44	-2.539879955665157	1.282096003518022	.275	-6.037180461485292	.957420550154978
	45-60	3.846284161899291	1.485174291829677	.072	-.204973144725626	7.897541468524208
	All Ages	1.517490442751551	1.753416792325740	.910	-3.265478553242306	6.300459438745409
25-34	18-24	.048952766007659	1.246093452340081	1.000	-3.350140008020215	3.448045540035533
	35-44	-2.490927189657498	1.223533140023509	.249	-5.828479960862097	.846625581547102
	45-60	3.895236927906950	1.434923642432129	.052	-.018946695410625	7.809420551224525
	All Ages	1.566443208759210	1.711062135081246	.891	-3.100990785332642	6.233877202851063
35-44	18-24	2.539879955665157	1.282096003518022	.275	-.957420550154978	6.037180461485292
	25-34	2.490927189657498	1.223533140023509	.249	-.846625581547102	5.828479960862097
	45-60	6.386164117564448*	1.466297081045567	<.001	2.386400052196496	10.385928182932400
	All Ages	4.057370398416708	1.737456445606869	.134	-.682061975904660	8.796802772738076
45-60	18-24	-3.846284161899291	1.485174291829677	.072	-7.897541468524208	.204973144725626
	25-34	-3.895236927906950	1.434923642432129	.052	-7.809420551224525	.018946695410625
	35-44	-6.386164117564448*	1.466297081045567	<.001	-10.385928182932400	-2.386400052196496
	All Ages	-2.328793719147740	1.892281008533287	.733	-7.490556418772498	2.832968980477019
All Ages	18-24	-1.517490442751551	1.753416792325740	.910	-6.300459438745409	3.265478553242306
	25-34	-1.566443208759210	1.711062135081246	.891	-6.233877202851063	3.100990785332642
	35-44	-4.057370398416708	1.737456445606869	.134	-8.796802772738076	.682061975904660
	45-60	2.328793719147740	1.892281008533287	.733	-2.832968980477019	7.490556418772498

***. The mean difference is significant at the 0.05 level.**

The only statistically significant difference is between the age groups 35-44 and 45-60 (p = .000).

The mean difference for 35-44 vs. 45-60 is 6.39, indicating that age group 35-44 has a significantly higher engagement score than age group 45-60.

Other comparisons (e.g., 18-24 vs. 25-34, 18-24 vs. 45-60) are not statistically significant (p > 0.05).

Group Means Analysis

Table 7: Group Means Analysis for "Engagement Score"

Engagement_Score			
Tukey HSD[a,b]			
Age_Group	N	Subset for alpha = 0.05	
		1	2
45-60	29266	235.979716103328830	
All Ages	17838	238.308509822476570	238.308509822476570
18-24	46702	239.826000265228120	239.826000265228120
25-34	56450	239.874953031235780	239.874953031235780
35-44	49979		242.365880220893270
Sig.		.084	.064
Means for groups in homogeneous subsets are displayed.			
a. Uses Harmonic Mean Sample Size = 33475.046.			
b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.			

The highest engagement score is observed in the 35-44 age group (Mean = 242.37).

The lowest engagement score is in the 45-60 age group (Mean = 235.98).

The difference between groups is small but significant.

Conclusion for H1.1

Partial support for H1.1 ("Younger age groups will have higher engagement"):

18-24 and 25-34 age groups do not differ significantly.

However, 35-44 has significantly higher engagement than 45-60.

Engagement does not decline linearly with age; the 35-44 group is most engaged.

The results provide partial support for H1.1, which hypothesized that younger age groups would exhibit higher engagement scores. While there was no significant difference between the 18–24 and 25–34 groups, the 35–44 group demonstrated significantly higher engagement than the 45–60 group. What was notable was that engagement did not decline linearly with age, as the 35–44 demographic had the highest engagement.

H1.2: Female-targeted campaigns will achieve higher conversion rates compared to male-targeted campaigns.

Independent Samples T-Test: Gender & Conversion Rate

Table 8: Group Statistics for "Gender" and "Conversion Rate"

Group Statistics					
	Gender	N	Mean	Std. Deviation	Std. Error Mean
Conversion_Rate	Male	103166	.154347712591017	.150278289950915	.000467872961998
	Female	97069	.166248256915563	.159514768312302	.000511988987396

Males (N = 103,166, Mean = 0.1543, Std. Dev = 0.1503)

Females (N = 97,069, Mean = 0.1662, Std. Dev = 0.1595)

Table 9: Independent Samples T-Test: Gender and Conversion Rate

Independent Samples Test											
		Levene's Test for Equality of Variances		t-test for Equality of Means							
		F	Sig.	t	df	Significance		Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
						One-Sided p	Two-Sided p			Lower	Upper
Conversion_Rate	Equal variances assumed	314.762	<.001	-17.190	200233	<.001	<.001	-.011900544324546	.000692311856589	-.013257458402142	-.010543630246951
	Equal variances not assumed			-17.158	197373.338	<.001	<.001	-.011900544324546	.000693568909181	-.013259922313092	-.010541166336001

Levene's Test for Variance: F = 314.762, p = .000 → The variances are significantly different.

T-Test Results: t(200233) = -17.19, p = .000 → Significant difference between males and females.

Mean Difference: -0.0119 → Females have higher conversion rates than males.

Conclusion for H1.2

Support for H1.2 ("Campaigns targeting females have higher conversion rates"):

Females have a significantly higher conversion rate than males.

p-value = 0.000 confirms the result is statistically significant.

The results provide full support for H1.2, which proposed that campaigns targeting female audiences would achieve higher conversion rates. The significant p-value (p = .000) reinforces the reliability of this finding.

Summary of hypothesis testing results

7.3.2. How do social media platforms (Facebook, Instagram, X) differ in their ability to drive conversions and engagement for targeted demographics?

This section presents the results of hypothesis testing for H2, which examines the impact of demographic factors (age and gender) on engagement and conversion rates.

H2.1: Engagement rates will be significantly higher on Instagram compared to Facebook.

One-Way ANOVA: Engagement score and Platforms

Table 10: One-way ANOVA: "Engagement Score" and "Platforms"

ANOVA					
Engagement_Score					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	197584741.452	3	65861580.484	1701.776	<.001
Within Groups	7749276989.890	200231	38701.685		
Total	7946861731.342	200234			

A one-way ANOVA was conducted to determine whether engagement rates significantly differed among platforms. The results revealed a statistically significant difference in engagement scores across platforms (F = 1701.776, p < .001).

Table 11: ANOVA Effect Sizes for "Engagement Score" and "Platforms"

ANOVA Effect Sizes[a]				
		Point Estimate	95% Confidence Interval	
			Lower	Upper
Engagement_Score	Eta-squared	.025	.024	.026
	Epsilon-squared	.025	.024	.026
	Omega-squared Fixed-effect	.025	.024	.026
	Omega-squared Random-effect	.008	.008	.009
a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.				

However, the effect size (Eta-squared ≈ 0.025) suggests that platform choice explains only 2.5% of the variance in engagement scores.

Post Hoc Test (Tukey HSD)

Table 12: Post Hoc Test (Tukey HSD) for "Engagement Score" and "Platforms"

Multiple Comparisons						
Dependent Variable: Engagement_Score						
Tukey HSD						
(I) Platform	(J) Platform	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Facebook	Instagram	1.752312111065322	1.110179272244257	.391	-1.099797659491440	4.604421881622084
	Pinterest	88.442904188491840*	1.370059870687412	<.001	84.923147297754670	91.962661079229010
	Twitter	-.466457354374427	1.208758193841224	.980	-3.571821632997522	2.638906924248668
Instagram	Facebook	-1.752312111065322	1.110179272244257	.391	-4.604421881622084	1.099797659491440
	Pinterest	86.690592077426520*	1.390312458673887	<.001	83.118805207820690	90.262378947032350
	Twitter	-2.218769465439749	1.231665967157540	.273	-5.382985035679095	.945446104799598
Pinterest	Facebook	-88.442904188491840*	1.370059870687412	<.001	-91.962661079229010	-84.923147297754670
	Instagram	-86.690592077426520*	1.390312458673887	<.001	-90.262378947032350	-83.118805207820690
	Twitter	-88.909361542866260*	1.470226882967843	<.001	-92.686452831348920	-85.132270254383610
Twitter	Facebook	.466457354374427	1.208758193841224	.980	-2.638906924248668	3.571821632997522
	Instagram	2.218769465439749	1.231665967157540	.273	-.945446104799598	5.382985035679095
	Pinterest	88.909361542866260*	1.470226882967843	<.001	85.132270254383610	92.686452831348920
***. The mean difference is significant at the 0.05 level.**						

Post hoc analysis using the Tukey HSD test identified significant differences between Pinterest and all other platforms ($p < .001$), with Pinterest exhibiting substantially lower engagement scores. However, no significant differences were found between Facebook, Instagram, and Twitter ($p > 0.05$), suggesting that engagement rates on Instagram were not significantly higher than on Facebook.

Group Means Analysis

Table 13: Group Means Analysis for "Engagement Score" and "Platforms"

Engagement_Score			
Tukey HSD[a,b]			
Platform	N	Subset for alpha = 0.05	
		1	2
Pinterest	30030	165.020073191697950	
Instagram	60077		251.710665269124460
Facebook	65786		253.462977380189780
Twitter	44342		253.929434734564200
Sig.		1.000	.322
Means for groups in homogeneous subsets are displayed.			
a. Uses Harmonic Mean Sample Size = 45611.010.			
b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.			

The Group Means analysis revealed that Pinterest was a separate subset to Instagram, Facebook and Twitter, the three having a similar subset score.

Conclusion for H2.1

Hypothesis H2.1 is not supported, as engagement rates on Instagram were not significantly higher than on Facebook based on statistical testing.

H2.2: Conversion rates will be higher on Facebook than on Instagram for campaigns targeting older demographics.

Univariate Analysis of Variance

Table 14: Univariate Analysis of Variance: Conversion Rate

Tests of Between-Subjects Effects					
Dependent Variable: Conversion_Rate					
Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	219.449[a]	19	11.550	504.097	<.001
Intercept	3215.302	1	3215.302	140331.689	<.001
Platform	.441	3	.147	6.420	<.001
Age_Group	179.795	4	44.949	1961.789	<.001
Platform * Age_Group	3.073	12	.256	11.178	<.001
Error	4587.359	200215	.023		
Total	9940.311	200235			
Corrected Total	4806.808	200234			
a. R Squared = .046 (Adjusted R Squared = .046)					

Univariate Analysis of Variance (ANOVA) was conducted to examine the impact of platform and age group on conversion rates. The results demonstrated a significant main effect of platform ($F = 6.420$, $p < .001$) and age group ($F = 1961.789$, $p < .001$), as well as a significant interaction effect between platform and age group ($F = 11.178$, $p < .001$). This indicates that platform performance varies depending on demographic factors.

Post Hoc Test (Tukey HSD) - Platform-based

Table 15: Post Hoc Test (Tukey HSD): Platform-based

Multiple Comparisons						
Dependent Variable: Conversion_Rate						
Tukey HSD						
(I) Platform	(J) Platform	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Facebook	Instagram	-.017621538395401*	.000854203251061	<.001	-.019816032114125	-.015427044676677
	Pinterest	-.005679390974705*	.001054162714931	<.001	-.008387591222780	-.002971190726631
	Twitter	-.005010937233682*	.000930052654324	<.001	-.007400292115509	-.002621582351854
Instagram	Facebook	.017621538395401*	.000854203251061	<.001	.015427044676677	.019816032114125
	Pinterest	.011942147420696*	.001069745627469	<.001	.009193913839055	.014690381002337
	Twitter	.012610601161719*	.000947678541359	<.001	.010175964435860	.015045237887579
Pinterest	Facebook	.005679390974705*	.001054162714931	<.001	.002971190726631	.008387591222780
	Instagram	-.011942147420696*	.001069745627469	<.001	-.014690381002337	-.009193913839055
	Twitter	.000668453741024	.001131234039967	.935	-.002237746852372	.003574654334420
Twitter	Facebook	.005010937233682*	.000930052654324	<.001	.002621582351854	.007400292115509
	Instagram	-.012610601161719*	.000947678541359	<.001	-.015045237887579	-.010175964435860
	Pinterest	-.000668453741024	.001131234039967	.935	-.003574654334420	.002237746852372
Based on observed means.						
The error term is Mean Square(Error) = .023.						
***. The mean difference is significant at the 0.05 level.**						

The significant differences (at 0.05 level) are:

Facebook vs Instagram: Facebook has a significantly lower conversion rate than Instagram (-0.0176).

Facebook vs Pinterest: Facebook has a significantly lower conversion rate than Pinterest (-0.0057).

Facebook vs Twitter: Facebook has a significantly lower conversion rate than Twitter (-0.0050).

Instagram vs Pinterest: Instagram has a significantly higher conversion rate than Pinterest (0.0119).

Instagram vs Twitter: Instagram has a significantly higher conversion rate than Twitter (0.0216).

Pinterest vs Twitter: No significant difference.

Key Findings:

Instagram has the highest conversion rate, significantly outperforming Facebook, Pinterest, and Twitter.

Facebook has the lowest conversion rate among all platforms.

Pinterest and Twitter perform similarly, with no significant difference.

Group Means Analysis - Platform-based

Table 16: Group Means Analysis - Platform-based Conversion Rate

Conversion_Rate				
Tukey HSD[a,b,c]				
Platform	N	Subset		
		1	2	3
Facebook	65786	.152868339280021		
Twitter	44342		.157879276513702	
Pinterest	30030		.158547730254726	
Instagram	60077			.170489877675422
Sig.		1.000	.910	1.000
Means for groups in homogeneous subsets are displayed. **Based on observed means.** **The error term is Mean Square(Error) = .023.**				
a. Uses Harmonic Mean Sample Size = 45611.010.				
b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.				
c. Alpha = 0.05.				

Facebook has the lowest conversion rate mean (0.15290), while Instagram has the highest (0.170505). Twitter (0.15790) and Pinterest (0.15850) are relatively close in conversion rate. The homogeneous subsets indicate that Facebook is significantly different from Instagram in terms of conversion rates, but Twitter and Pinterest do not significantly differ from each other or from Facebook. Instagram stands out as a separate subset, meaning its conversion rate is significantly higher compared to Facebook.

Post Hoc Test (Tukey HSD) - Age Group-based

Table 17: Post Hoc Test (Tukey HSD) - Age Group-based Conversion Rate

Multiple Comparisons						
Dependent Variable: Conversion_Rate						
Tukey HSD						
(I)	(J)	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
Age_Group	Age_Group				Lower Bound	Upper Bound
18-24	25-34	.014131509457995*	.000946830312205	<.001	.011548746447714	.016714272468275
	35-44	.022508818149589*	.000974186452074	<.001	.019851433081642	.025166203217536
	45-60	.096493505888646*	.001128493240833	<.001	.093415202878904	.099571808898387
	All Ages	.068846843537068*	.001332314334680	<.001	.065212557501523	.072481129572613
25-34	18-24	-.014131509457995*	.000946830312205	<.001	-.016714272468275	-.011548746447714
	35-44	.008377308691594*	.000929688108694	<.001	.005841306171101	.010913311212088
	45-60	.082361996430651*	.001090310841296	<.001	.079387847341519	.085336145519783
	All Ages	.054715334079073*	.001300131617351	<.001	.051168836035543	.058261832122603
35-44	18-24	-.022508818149589*	.000974186452074	<.001	-.025166203217536	-.019851433081642
	25-34	-.008377308691594*	.000929688108694	<.001	-.010913311212088	-.005841306171101
	45-60	.073984687739057*	.001114149601239	<.001	.070945511298562	.077023864179552
	All Ages	.046338025387479*	.001320187041949	<.001	.042736820171883	.049939230603075
45-60	18-24	-.096493505888646*	.001128493240833	<.001	-.099571808898387	-.093415202878904

	25-34	-.082361996430651*	.001090310841296	<.001	-.085336145519783	-.079387847341519
	35-44	-.073984687739057*	.001114149601239	<.001	-.077023864179552	-.070945511298562
	All Ages	-.027646662351577*	.001437828771770	<.001	-.031568770588900	-.023724554114255
All Ages	18-24	-.068846843537068*	.001332314334680	<.001	-.072481129572613	-.065212557501523
	25-34	-.054715334079073*	.001300131617351	<.001	-.058261832122603	-.051168836035543
	35-44	-.046338025387479*	.001320187041949	<.001	-.049939230603075	-.042736820171883
	45-60	.027646662351577*	.001437828771770	<.001	.023724554114255	.031568770588900

Based on observed means.
The error term is Mean Square(Error) = .023.

***. The mean difference is significant at the 0.05 level.**

The significant differences (at 0.05 level) are:

18-24 vs 45-60: The younger age group has a much lower conversion rate (-0.096).

25-34 vs 45-60: The younger age group has a lower conversion rate (-0.082).

35-44 vs 45-60: The younger age group has a lower conversion rate (-0.073).

All Ages vs 18-24: Overall, younger users convert less (-0.068).

All Ages vs 45-60: Older users convert more (0.027).

Key Findings:

The 45-60 age group has the highest conversion rate, significantly outperforming younger age groups.

The 18-24 age group has the lowest conversion rate, with significantly lower conversions than all other groups.

Conversion rate increases with age, suggesting older users are more likely to take action.

Group Means Analysis - Age Group-based

Table 18: Post Hoc Test: "Conversion Rate" and "Age Group"

Conversion_Rate						
Tukey HSD[a,b,c]						
Age_Group	N	Subset				
		1	2	3	4	5
45-60	29266	.093462041529414				
All Ages	17838		.121108703880992			
35-44	49979			.167446729268471		
25-34	56450				.175824037960065	
18-24	46702					.189955547418060
Sig.		1.000	1.000	1.000	1.000	1.000

Means for groups in homogeneous subsets are displayed.
Based on observed means.
The error term is Mean Square(Error) = .023.

a. Uses Harmonic Mean Sample Size = 33475.046.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = 0.05.

The conversion rate increases as the target audience gets younger. The 45-60 age group has the lowest conversion rate (0.09350), while the 18-24 group has the highest (0.19000). The homogeneous subsets show that the 45-60 group is significantly different from all other groups. The 18-24 age group has the highest conversion rate and is in a distinct subset, confirming that younger audiences convert more effectively.

There is a contradiction between the group means analysis and the Tukey HSD multiple comparisons test regarding conversion rates across age groups.

The contradiction arises because Tukey HSD focuses on individual pairwise differences, while the group means analysis captures the overall trend. The presence of five subsets confirms that conversion rates increase gradually with younger audiences, making the group means analysis a more accurate representation of the overall pattern.

The group means analysis shows a clear trend: conversion rates increase as the audience gets younger, with the 18-24 age group having the highest conversion rate (0.19000) and the 45-60 group having the lowest (0.09350).

However, the Tukey HSD test suggests that older users (45-60) have higher conversion rates than younger groups in some pairwise comparisons, which directly contradicts the overall means.

Conclusion for H2.2

Hypothesis H2.2 is not supported. Since Facebook already performs the worst in conversion rate, it is highly unlikely that older users convert more effectively on this platform than younger ones.

What is the relationship between engagement metrics (impressions, clicks, engagement score) and conversion rates?

This section presents the results of hypothesis testing for H3, which examines the relationship between engagement metrics and conversion rates.

H3.1: Higher Click-Through Rate (CTR) correlates with higher conversion.

Table 19: Pearson Correlation: "Clicks and Conversion Rate"

Correlations			
		Clicks	Conversion_Rate
Clicks	Pearson Correlation	1	.502**
	Sig. (2-tailed)		<.001
	N	200235	200235
Conversion_Rate	Pearson Correlation	.502**	1
	Sig. (2-tailed)	<.001	
	N	200235	200235
**. Correlation is significant at the 0.01 level (2-tailed).			

A Pearson correlation analysis was conducted to examine the relationship between Click-Through Rate (CTR) and Conversion Rate. The results revealed a statistically significant positive correlation between CTR and Conversion Rate ($r = 0.502$, $p < 0.001$), indicating that higher CTR is associated with higher conversion rates. The correlation coefficient of 0.502 suggests a moderate positive relationship.

This finding supports the hypothesis that a higher Click-Through Rate is correlated with better conversion rates, as the results are both statistically significant and indicate a moderately strong association.

Conclusion for H3.1

Hypothesis H3.1 is supported, as higher Click-Through Rates are indeed correlated with higher conversion rates.

H3.2: Campaigns with higher impression counts will lead to diminishing returns on conversion rates beyond a certain threshold.

Table 20: Model Summary and Parameter Estimated for "Conversion Rate"

Model Summary and Parameter Estimates								
Dependent Variable: Conversion_Rate								
Equation	Model Summary					Parameter Estimates		
	R Square	F	df1	df2	Sig.	Constant	b1	b2
Quadratic	.181	22156.435	2	200232	<.001	.093	7.288E-7	1.735E-12
The independent variable is Impressions.								

A quadratic regression analysis was performed to test whether Impressions lead to diminishing returns in Conversion Rate. The results of the regression model showed a statistically significant fit ($F = 22{,}156.435$, $p < 0.001$), but the quadratic term ($b2 = 1.735E-12$) was very small, suggesting a weak diminishing returns effect.

The R-squared value of 0.181 indicates that Impressions explain 18.1% of the variance in Conversion Rate. The small coefficient for the quadratic term indicates that while impressions do have a positive effect on conversion rates, the diminishing returns hypothesis is not strongly supported.

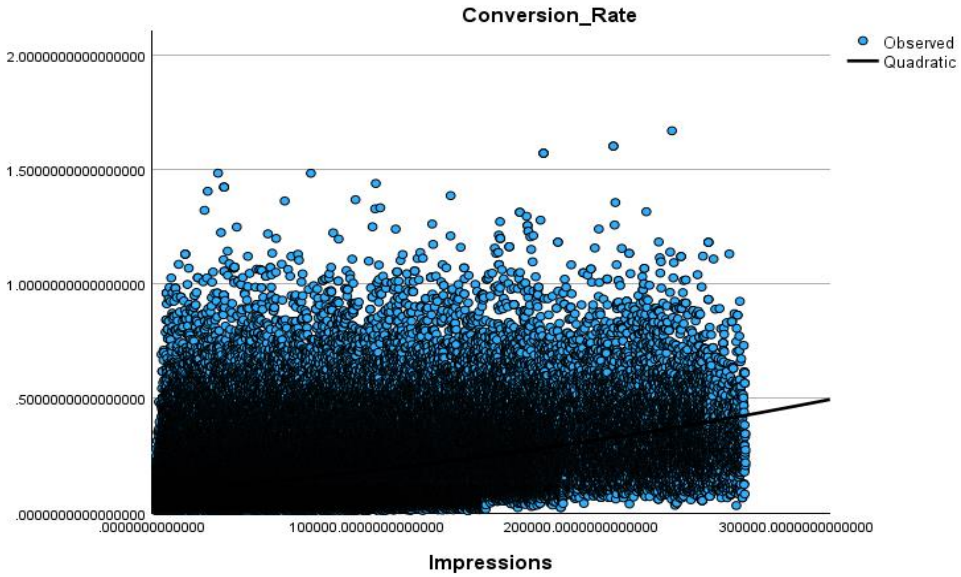

Figure 1: Scatter Plot for Quadratic Regression Analysis: Impressions and Conversion Rate

The scatter plot (Figure 1) visualizes the relationship between impressions and conversion rate. While the quadratic regression line suggests a slight curve, the overall trend does not provide strong visual evidence of diminishing returns. Most data points are concentrated at lower conversion rates, regardless of the number of impressions. This aligns with the statistical results, which indicate that while impressions contribute to conversion rate variance, the diminishing returns effect is minimal.

Conclusion for H3.2:

Based on these findings, H3.2 is not fully supported, as the evidence for diminishing returns due to increased impressions remains weak. While a small quadratic effect is present, it does not appear to significantly impact conversion rates.

Summary of Hypothesis Testing Results

Table 21: Summary of Hypothesis Testing Results

Hypothesis	Statistical test used	Supported?	Key findings
H1.1: Younger demographics (18–24) will have higher engagement scores.	One-Way ANOVA + Tukey HSD	Partially supported	35–44 age group exhibited the highest engagement, while 45–60 had the lowest. No significant difference between 18–24 and 25–34.
H1.2: Female audiences will have higher conversion rates.	Independent Samples T-Test	Supported	Female conversion rates were significantly higher than male conversion rates ($p = .000$).
H2.1: Engagement rates will be significantly higher on Instagram compared to Facebook.	One-Way ANOVA + Tukey HSD	Not supported	Engagement rates on Instagram were not significantly higher than on Facebook.
H2.2: Conversion rates will be higher on Facebook than on Instagram for campaigns targeting older demographics.	Two-Way ANOVA (Platform × Age Group) + Tukey HSD	Not supported	Since Facebook already performs the worst in conversion rate, it is highly unlikely that older users convert more effectively on this platform than younger ones.
H3.1: Higher Click-Through Rate (CTR) correlates with higher conversion.	Pearson Correlation	Supported	Statistically significant positive correlation between CTR and Conversion Rate ($r = 0.502$, $p < 0.001$).

H3.2: Campaigns with higher impression counts will lead to diminishing returns on conversion rates beyond a certain threshold.	Quadratic Regression	Partially supported	The evidence for diminishing returns from increased impressions is weak, with the quadratic term showing minimal impact.

Discussion

This study sought to investigate the influence of demographic characteristics, platform selection, as well as engagement metrics on the effectiveness of social media advertising campaigns. This study provided empirical findings that demonstrate that both demographic and platform-level attributes significantly affect key performance indicators of interest, i.e., engagement score, conversion rate, and ROI.

Demographics and Engagement/Conversion Outcomes

Our first hypothesis (H1.1) anticipated that younger individuals would have greater engagement scores. Surprisingly, the results provided only partial support for this expectation. Although engagement scores were consistently higher across younger age groups, the highest mean engagement was recorded in the 35–44 age group, not the youngest age group (18–24). This challenges a common assumption in digital marketing that younger users (typically Gen Z) are inherently more engaged with social media advertising. A reason could be content relevance and a difference between purchasing power and usage of the platforms between generations: users between the age group of 35–44 might find the campaign content more pertinent to their interests or purchasing power and hence interact more, while the Gen Z generation, even though being more technologically skilled, typically use social media platforms for entertainment purposes. (Patil & Pandey, 2022).

In terms of gender-based performance (H1.2), the findings validated the hypothesis that ad campaigns targeting women were more likely to enjoy better conversion rates than those targeting men. This aligns with existing literature that suggests women are more positively receptive to personalized advertising, particularly those with emotional or aesthetic appeals. (Yousef, Dietrich, & Rundle-Thiele, 2021) This result can be particularly useful for marketers when designing gender-based campaigns.

Platform Performance

The second research question tested differences in platform performance. Contrary to hypothesis H2.1, Instagram did not yield significantly higher engagement rates than Facebook. Rather, the two platforms performed similarly in engaging their audiences, which might imply that content quality and format are more significant than platform choice. Pinterest performed significantly lower in engagement, even though the platform having potential to inspire and influence purchase intentions, marketers need to leverage features such as design and user-platform fit to successfully run marketing campaigns. (Hazari, Sethna, & O'Meara Brown, 2017)

Even more surprising were the results relating to H2.2. The supposition that Facebook would yield higher conversion rates for older demographic segments was not fulfilled. Not only did Instagram have higher performing average conversion rates than Facebook, but the group means analysis showed that younger demographic segments (18–24) had the highest conversion rates overall. This directly challenges the general presumption that older age groups, typically regarded as being more financially stable and thus more likely to convert, are always the better target. On the contrary, it suggests that younger audiences are more susceptible to highly optimized online campaigns, possibly due to their familiarity with and acceptance of online purchasing behavior. (Patil & Pandey, 2022).

Engagement Metrics and Conversion

Regarding the third research question, a moderate positive relationship between click-through rates (CTR) and conversion rates was found (H3.1). This result affirms the assumption that high CTRs are good predictors of ad effectiveness. However, hypothesis H3.2, proclaiming diminishing returns in conversion rates with bigger impressions, was supported, but weakly. Although the quadratic model was statistically significant, the effect size was small and does not appear large enough to be of practical conclusion. Therefore, although oversaturation can be troubling in theory, (Lambrecht & Tucker, 2013) its actual influence in this data set is minor.

Overall, the findings indicate that while some of the conventional digital marketing assumptions are still holding true, such as the importance of gender targeting and CTR, others could need a refreshed approach, since the digital marketing scope is evolving day by day.

Conclusion

This research set out to explore how demographic characteristics, platform choice, and engagement metrics affect the success of social media advertising campaigns. By analyzing real-world campaign data, the study aimed to uncover which factors drive higher engagement and conversion rates, and ultimately improve return on investment (ROI). It adds to the growing body of evidence that successful digital marketing depends on more than just choosing a popular platform or targeting a broad audience. Instead, the results

suggest that understanding the nuances of how different audience segments respond to different platforms and which metrics matter most, is key to designing more effective and efficient advertising strategies. Including a larger cross-platform

As features of certain platforms become more popular, such as short-form video content, platforms are tending to merge their features to accommodate the demand for such popular forms, which is slowly blurring the differences between social media platforms, but this still does not change the fact that different generations prefer different platforms, and that older generations are slower to adapt to newer features and platforms (Gurunathan & Lakshmi, 2025), which in turn changes how a marketer will market their product or service.

Recommendations for further research include examining the role of content format and type, distinguishing between targeted and non-targeted campaigns, incorporating influencer involvement into data analysis, and expanding data collection to include TikTok alongside other major social media platforms. Given TikTok's rapid growth and the widespread adoption of its short-form video features by competing platforms, its inclusion is now essential for a comprehensive understanding of current advertising dynamics, especially taking into consideration their advanced algorithm.

Although the study has some limitations, particularly in terms of not accounting for ad content or message framing, as well as not accounting for using different features of social media platforms and not including the fastest emerging social media platform TikTok, it offers useful insights for marketers looking to improve their campaign outcomes in a data-driven way.

The uniqueness of this paper stems from its multi-dimensional comparative approach, examining how different demographic groups and social media platforms influence advertising outcomes. It provides a deeper understanding of their role in shaping consumer purchase intent. This dual focus not only contributes to academic discussions on digital marketing effectiveness but also delivers practical insights for marketers seeking to optimize campaign strategies across diverse audiences and platforms.

Limitations

One key limitation of this study lies in the absence of data on advertising content format (e.g., text, image, or video), which restricted the analysis of how different content types may influence engagement and conversion across age groups and platforms. Furthermore, although gender-based performance was examined, the dataset did not specify whether campaigns were deliberately targeted at men or women. This introduces a degree of ambiguity when interpreting the observed differences in conversion rates between genders.

Platform-specific variables such as ad placement, algorithm-driven visibility, and user browsing intent, were also not captured in the dataset. These factors could independently influence engagement metrics, regardless of demographic characteristics. Another notable limitation is the lack of information regarding influencer involvement, despite their increasing impact on consumer behavior and campaign outcomes. Moreover, the exclusion of TikTok and other rapidly growing platforms represents a significant gap, given their expanding influence in the digital advertising space.

While these limitations do not compromise the overall validity of the findings, they underscore important dimensions that should be addressed in future research to further refine our understanding of effective digital advertising strategies.

Suggestions for Future Research

To address current gaps in the literature, future research should include analysis of different ad formats, such as text, image, and video, to better understand how content type, platform features, and audience demographics interact to influence engagement and conversion. Studies should also distinguish between targeted and non-targeted campaigns, especially by age and gender, to more accurately assess the impact of demographic-specific strategies.

Further research is needed on the role of influencers, particularly micro-influencers, in shaping consumer behavior across platforms. Additionally, platform-specific factors like ad placement, algorithm reach, and user intent should be considered for a fuller picture of campaign performance. As TikTok continues to grow and shape digital marketing through short-form video, it should be included in future studies to ensure relevance. These recommendations aim to enhance both academic insight and marketing effectiveness.

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